LEARNER GUIDE

Conversion graphs as models

Straight Lines and Connected Representations. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A straight line can do more than describe a rule — it can act as a converter. This lesson uses a line to translate one quantity into another: miles into kilometres, pounds into euros, degrees Celsius into Fahrenheit. The key skill is reading the graph BOTH ways — go up from one axis to the line, then across to the other, or across-and-down to convert back. You will read conversions off a line, drag a point to convert any value, spot the common slips, and use a pounds-to-euros graph to sort out your holiday money. The question you are exploring is: how can a line translate one quantity into another?

What you are learning to notice

- That one straight line can convert between two quantities.
- That you read it both ways: up-and-across one direction, across-and-down the other.
- That the gradient is the conversion rate — how many of one unit per one of the other.
- That some conversion lines pass through the origin (0 miles = 0 km) and some do NOT (0°C = 32°F), so you must read to the line rather than just multiplying.

Moving through it, one step at a time

In the Conversion Graph Translator, read a value up to the line and across to the other axis, then type the answer. In the Two-way reader, drag the point along the line — dashed guides drop to both axes, showing the matching pair, and you can switch between miles/km, £/€ and °C/°F. In Read it right?, decide whether a reading is correct or a slip. Then use the Holiday money graph to convert pounds and euros both ways. There is no rush; use the coordinate grid on the Scratchpad to draw a conversion line and read it for yourself.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (reading coordinates from a graph, reading a value up to a line, and understanding the gradient as a rate), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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